

Shubham Kulkarni

AI Engineer — Computer Vision — Generative AI — MLOps — Edge AI

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PROFESSIONAL SUMMARY

Production-focused AI Engineer specializing in Computer Vision, Generative AI, and MLOps. Proven track record of architecting and deploying scalable, end-to-end intelligent systems that process over 150,000 events daily with 95% accuracy. Expertise spans edge AI deployment on NVIDIA Jetson architectures, low-latency concurrent inference (sub-40ms), and multi-modal Retrieval-Augmented Generation (RAG) pipelines. Recognized nationally as an All India Rank 1 (AIR 1) winner at the Smart India Hackathon for developing high-impact, real-world AI solutions. Driven to engineer highly optimized, autonomous systems that bridge advanced machine learning models with resilient enterprise infrastructure.

TECHNICAL SKILLS

- AI/ML & Vision:** Deep Learning, Large Language Models (LLMs), YOLOv5/v8, RT-DETR, ByteTrack, OSNet, ResNet, PaddleOCR (PP-OCRv4), Computer Vision, NLP.
- Edge AI & Hardware:** NVIDIA Jetson (Orin/NX/Xavier/Nano), TensorRT (FP16/INT8), CUDA, GStreamer, Hardware-Accelerated Video Decoding (nvv412decoder), Raspberry Pi, Rockchip.
- MLOps & Backend:** Python, FastAPI, Django, Flask, AsyncIO, REST APIs, Microservices, Docker, CI/CD, Git, Multi-threading, Bounded Queues (Deque), WebSockets (SSE).
- Generative AI:** Retrieval-Augmented Generation (RAG), Prompt Engineering, LangChain, FAISS, Vector Databases, Semantic Chunking, Model Context Protocol (MCP), Multi-Agent Systems.
- Databases & Cloud:** PostgreSQL, Redis, AWS, Microsoft Azure, Google Cloud Platform (GCP), SQL.

PROFESSIONAL EXPERIENCE

Arya Omnitalk Wireless Solutions Pvt. Ltd.

Software Engineer - AI/ML

Pune, Maharashtra

Aug 2025 – Present

- Edge-Native Pipeline Architecture:** Spearheaded the edge-native architecture for enterprise Intelligent Transportation Systems (VIDES, ATMS), scaling concurrent multi-model pipelines to process 50,000+ daily real-time events with 95% enforcement accuracy.
- Hardware Acceleration & Latency Optimization:** Engineered zero-latency video ingestion by bypassing host CPU bottlenecks via GStreamer and nvv412decoder, achieving a strict sub-40ms processing lock across 6 concurrent HD RTSP streams on NVIDIA Jetson.
- Model Compilation & Tracking:** Accelerated deep learning inference by compiling ONNX models into TensorRT FP16 engines to maximize Ampere GPU throughput. Orchestrated YOLOv8, RT-DETR, and ByteTrack for multi-lane vehicle taxonomy and spatial trajectory hazard detection.
- Asynchronous System Design:** Architected highly resilient asynchronous data relays, decoupling heavy AI inference workloads from UI layers via PostgreSQL thread pools and sub-150ms WebSocket (SSE) broadcasts for instant operator alerts.
- Automated Data Extraction:** Automated highway toll auditing by engineering advanced OCR pipelines using PaddleOCR consensus logic, successfully eliminating manual reporting workflows and ensuring 100% data integrity.
- Infrastructure Reliability:** Ensured 99.9% edge-node uptime by implementing robust CI/CD pipelines, automated heartbeat monitors, and exponential backoff retry protocols for seamless offline network recovery.

MathWorks

Artificial Intelligence Intern

Remote

May 2023 – Sep 2023

- R&D Algorithms:** Engineered predictive models and statistical algorithms using MATLAB, establishing a rigorous foundation in deep learning and computer vision optimization techniques.

HIGH-IMPACT ENGINEERING PROJECTS

Document AI + Retrieval-Augmented Generation (RAG) Pipeline

Mar 2026 – Present

Python — FastAPI — FAISS — Mistral — Ollama — LangChain — AsyncIO

- Architecture:** Architected a secure, locally-hosted RAG system utilizing FastAPI and AsyncIO to process complex enterprise documents (PDFs, Images, Web URLs) with absolute data privacy and sub-2 second latency.
- Hybrid Search:** Engineered a high-precision retrieval engine merging FAISS-based dense vector search with BM25 sparse keyword search via Reciprocal Rank Fusion (RRF), significantly reducing LLM hallucinations.
- Knowledge Graphs:** Integrated interactive Knowledge Graphs for automated entity extraction and implemented dynamic routing between local LLMs (Mistral 7B) and cloud APIs based on query complexity.

Vehicle Forensic Matching System (RAC)

Dec 2025 – Mar 2026

PyTorch — YOLOv5 — FAISS — FastAPI — OSNet-AIN — Docker

- **System Design:** Designed a stateless, microservice-based vehicle re-identification platform for real-time forensic video analysis across non-overlapping camera networks with zero database dependencies.
- **Deep Embeddings:** Generated 512-dimensional deep feature embeddings using OSNet-AIN and localized vehicles via YOLO, ensuring high-fidelity matches across varied lighting and extreme occlusion conditions.
- **Vector Search:** Optimized spatial retrieval speed utilizing FAISS IVF indexing, achieving sub-100ms vector search latency across 150,000+ surveillance records within a scalable Dockerized environment.

JalTantra (Smart Irrigation & AgTech Platform) – SPPU Startup Olympiad Rank 2 Dec 2024 – May 2025

Python — IoT — Computer Vision — Machine Learning — GenAI

- **Platform Development:** Spearheaded an AI-powered smart agriculture platform integrating precision IoT irrigation logic, predictive yield modeling, and real-time computer vision for automated crop disease detection.
- **Execution:** Achieved a 30% improvement in water efficiency through sensor-based automation and deployed a multi-lingual AI advisory chatbot (Hindi, Marathi, English) for instant farmer support.
- **Impact:** Pitched and demonstrated the live platform at the SPPU Startup Olympiad 2025, securing University Rank 2 for high commercial viability and social impact.

Alumni Management Portal (Enterprise Full-Stack Application)

Django DRF — PostgreSQL — Flask — WebSockets — React/Vanilla JS ES6

- **Architecture:** Architected a Hybrid Monolith application to replace legacy university systems, supporting high-concurrency mentorship pairing and event management workflows.
- **Optimization:** Slashed database query latency from 3,200ms to 12ms utilizing composite indexing and query profiling, and engineered a custom WebSocket subscriber queue to resolve JWT refresh race conditions.

EDUCATION

Savitribai Phule Pune University (SPPU) Dec 2021 – Mar 2025

Bachelor of Engineering (B.E.) in Electronics and Telecommunication First Class

- **Key Projects:** Engineered the officially adopted **Alumni Management Portal**, and developed the **AI E-Waste Management System** (Winner, AIR 1 at Smart India Hackathon 2023).

Residential Arts, Commerce and Science Junior College Jun 2019 – Mar 2021

Higher Secondary Certificate (12th Grade, Science) 87.50% (First Class with Distinction)

- **Specialization:** Computer Science (C++, VB, Binary, Basic Electronics).

ACHIEVEMENTS & CERTIFICATIONS

- **All India Rank 1 (AIR 1) – Smart India Hackathon (SIH) 2023:** Won the National Championship in the software edition for engineering an intelligent e-waste routing and classification system.
- **University Rank 2 – SPPU Startup Olympiad 2025:** Recognized for "JalTantra," demonstrating high commercial viability in the AgTech sector.
- **Deploy Multi-Agent Architectures** – Google (Apr 2026) — **Engineer AI Agents (ADK)** – Google Cloud (Feb 2025)
- **Introduction to Machine Learning Concepts** – Microsoft (Apr 2026)